From Local Curvature to the Weil Functional:

An Explicit Construction

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Abstract

Building on the curvature decomposition established in [1], we construct an explicit family of admissible test functions $g^*_{\sigma,\varepsilon} \in S_{\mathrm{ad}}$ for the Weil explicit formula and identify the corresponding prime-side contribution with the local curvature $H_{\mathrm{local}}(\sigma, \kappa)$ studied in [1]. A renormalised family of prime weights $c_p^{\mathrm{ren}} = f_p^{1/2} > 0$ is introduced, and the associated diagonal energy $D = \sum_p (c_p^{\mathrm{ren}})^2 \approx 9.471$ is shown to converge. The renormalised local curvature is shown to be of lower order than $(\log \kappa)^2$ and to exhibit a sawtooth structure governed, on average, by the Mertens constant. No hypothetical input is used: no Riemann Hypothesis, no Hilbert–Pólya postulate, no GUE conjecture.

What is proved. The antisymmetrised Gaussian family $g^*_{\sigma,\varepsilon}$ belongs to the admissible class S_{ad} used in Lagarias [3] (Lemma 2.1). The renormalised prime weight $c_p^{\mathrm{ren}} := \big(V_p(\frac{1}{2}) - \frac{4(\log p)^2}{p}\big)^{1/2}$ is strictly positive (Lemma 4.1), and the diagonal energy series $D = \sum_p (c_p^{\mathrm{ren}})^2$ converges absolutely (Lemma 4.2).

What is conditional. The prime-side identity

$$W(g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}) \;=\; Z(g^*_{\sigma,\varepsilon}) \;-\; H_{\mathrm{local}}(\sigma, \kappa) \;+\; R(\sigma, \varepsilon, \kappa)$$

holds with $R(\sigma, \varepsilon, \kappa) \to 0$ as $\varepsilon \to 0$ at fixed (σ, κ) (Proposition 3.1), conditional on the standard remainder bookkeeping for the Weil explicit formula applied to antisymmetric Schwartz test functions. The remainder is not made uniform in κ.

What is numerical. At reference parameters $(\varepsilon, N) = (0.05, 100)$, the diagonal energy evaluates to $D \approx 9.471$ on primes below 10^4, with tail bound < 0.009. Direct grid evaluation of the renormalised quantities at $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$ shows that $Z_{\mathrm{ren}} \geq H^{\mathrm{ren}}_{\mathrm{local}}$ holds at every tested point except $\kappa = 20$, where $Z_{\mathrm{ren}} - H^{\mathrm{ren}}_{\mathrm{local}} = -0.988$, with the ratio $Z_{\mathrm{ren}}/H^{\mathrm{ren}}_{\mathrm{local}}$ growing from 1.5 at $\kappa = 29$ to over 10 at $\kappa = 1000$. No claim is made about κ outside the eight tested points.

What remains open. Two questions separate the present construction from a full prime-side derivation of Weil positivity for this family (§ 7). Problem 7.1 (*Spectral inequality*) asks whether $Z(g^*_{1/2,\varepsilon}) \geq H_{\mathrm{local}}(\frac{1}{2}, \kappa)$ holds unconditionally; an unconditional proof for this family alone is necessary, but not sufficient, for the Riemann Hypothesis in the absence of a density argument in S_{ad}. Problem 7.2 (*Off-diagonal control*) asks for an unconditional bound on the cross-terms $\sum_\gamma \cos(\gamma \log(p/q))$ for $p \neq q$, which would upgrade the formal asymptotic $Z(g^*) \sim D \log \kappa \log\log \kappa$ to a proved statement.

Structure. The formal core in §§ 2–5 is non-circular, with proved and conditional components explicitly separated. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The construction parameter $\sigma > 0$ is evaluated at $\sigma = \frac{1}{2}$ in the renormalised analysis unless stated otherwise.

Global parameters. The reference values are $\kappa \geq 2$ (prime cutoff, varied across the comparison grid), $\varepsilon = 0.05$ (Gaussian smoothing parameter), and $N = 100$ (number of Riemann ordinates γ_k used in the zero-sum evaluations of § 6). Values are fixed within each statement and explicitly varied otherwise.

Main objects. The local curvature, imported from [1], is

$$H_{\mathrm{local}}(\sigma, \kappa) \;=\; \psi_1(\sigma) \;+\; \sum_{p \leq \kappa} V_p(\sigma), \qquad V_p(\sigma) \;=\; 4(\log p)^2 \, \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}.$$

The Gaussian bump is $\varphi_\varepsilon(x) = (\varepsilon\sqrt{2\pi})^{-1} e^{-x^2/(2\varepsilon^2)}$. The raw weight at level σ is $c_{p,\sigma} = V_p(\sigma)^{1/2} \cdot p^{1/4}/(\log p)^{1/2}$, and the antisymmetrised admissible test function $g^*_{\sigma,\varepsilon} \in S_{\mathrm{ad}}$ is constructed in § 2. The Weil quadratic functional is denoted W, and $Z(g) = \sum_\rho |\widehat{g}(\rho)|^2$ is the corresponding zero-side sum over the non-trivial zeros of ζ. At $\sigma = \frac{1}{2}$ the renormalised prime weight is $c_p^{\mathrm{ren}} = f_p^{1/2}$, where $f_p = V_p(\frac{1}{2}) - 4(\log p)^2/p$, and the diagonal energy is $D = \sum_p (c_p^{\mathrm{ren}})^2$. The renormalised local curvature is $H^{\mathrm{ren}}_{\mathrm{local}}(\frac{1}{2}, \kappa) = H_{\mathrm{local}}(\frac{1}{2}, \kappa) - 2(\log \kappa)^2$.

Not to be confused with. The diagonal energy $D \approx 9.471$ defined here, characteristic of the renormalised prime-side construction of this paper, must not be confused with the trace-formula diagonal D_{SEL} appearing in later work in the series, which is defined in a different operator context and takes a different numerical value. The renormalised weight $c_p^{\mathrm{ren}} = f_p^{1/2}$ used in the Weil functional differs in normalisation from the unrenormalised weight $c_{p,\sigma}$ used in the construction of $g^*_{\sigma,\varepsilon}$.

1 Introduction

1.1 Background and Motivation

This paper is the second in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The central organising principle of the series is that prime numbers mediate coupling between zero ordinates, and that explicit finite-cutoff constructions on the prime side carry arithmetic information accessible by classical means.

Paper 1 [1] introduced the curvature field

$$H_\xi(\sigma,t) \;:=\; \frac{d^2}{d\sigma^2}\,\log|\xi(\sigma+it)|^2 \;=\; H_{\mathrm{local}}(\sigma,\kappa) \;+\; H_{\mathrm{dual}}(\sigma,t,\kappa),$$

where $H_{\mathrm{dual}} = H_\xi - H_{\mathrm{local}}$ is the spectral remainder encoding zero contributions, and where the truncated local curvature

$$H_{\mathrm{local}}(\sigma,\kappa) \;=\; \psi_1(\sigma) \;+\; \sum_{p\le\kappa} V_p(\sigma)$$

contains the archimedean term and the finite-place contributions. The prime-side pieces satisfy $V_p(\sigma)\ge 0$, and on the critical line one has $H_{\mathrm{local}}(\frac{1}{2},\kappa) \asymp (\log\kappa)^2$.

The present paper addresses a concrete part of the central question raised in [1, § 7]: whether the local positivity and critical divergence of H_{local} can be organised into an explicit admissible test-function construction whose prime-side term in the Weil functional matches the local curvature. We answer this in the affirmative on the prime side by constructing an explicit family $g^*_{\sigma,\varepsilon} \in S_{\mathrm{ad}}$ and identifying the corresponding prime-side contribution with $H_{\mathrm{local}}(\sigma,\kappa)$ up to a smoothing remainder that vanishes as $\varepsilon\to 0$. The remaining open step — the spectral (zero-side) inequality on the critical line — is then isolated as a single RH-strength question in § 7.

The construction $g^*_{\sigma,\varepsilon} \in S_{\mathrm{ad}}$ developed here should be distinguished from the established operator-theoretic programmes addressing the Riemann Hypothesis. It is not a Hilbert–Pólya spectral realisation in the sense of Connes [9] or Berry–Keating [10], nor a de Branges structure-function construction [11]; it is an explicit antisymmetrised Gaussian smearing of prime delta masses with arithmetic weights, closest in spirit to Hilbert-space reformulations of the explicit formula such as those of Burnol [12]. The local curvature $H_{\mathrm{local}}(\sigma,\kappa)$ is, in the classical dictionary of analytic number theory [13], the truncated prime-sum side of the Weil explicit formula together with the archimedean term $\psi_1(\sigma)$; the curvature label reflects the second-derivative origin of $V_p(\sigma)$ in $\log|\xi|^2$ identified in [1] and is not used in any further geometric sense. The terms "prime–zero coupling" and "curvature field" used informally in the series are translations of the explicit-formula duality and of $\partial_\sigma^2 \log|\xi|^2$ respectively, not new mathematical structures.

The construction is set within the Lagarias formulation of Weil positivity [3], which characterises the Riemann Hypothesis as positivity of the Weil functional $W(f*\check{f})$ on a suitable class of test functions, building on the Weil explicit formula [4] and the Li positivity criterion [5]. The renormalised diagonal scale $D = \sum_p (c_p^{\mathrm{ren}})^2$ encountered in § 4 is closely related, via the prime number theorem and Mertens' classical estimates [6], to the average behaviour of the local curvature; this connection is made precise in § 5.

1.2 Main Results

The paper contains three main components.

The first is the explicit construction of the test family $g^*_{\sigma,\varepsilon}$ and the proof that it lies in the admissible class S_{ad} used by Lagarias [3]. The construction proceeds in two steps: a Gaussian smearing of delta masses at $\log p$ with weights $c_{p,\sigma}$, followed by antisymmetrisation enforcing the vanishing condition at the origin (§ 2). The admissibility lemma (Lemma 2.1) is a direct verification.

The second is the prime-side identity (Proposition 3.1), established conditionally on the standard remainder bookkeeping for the Weil explicit formula applied to antisymmetric Schwartz test functions:

$$W(g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}) \;=\; Z(g^*_{\sigma,\varepsilon}) \;-\; H_{\mathrm{local}}(\sigma,\kappa) \;+\; R(\sigma,\varepsilon,\kappa), \qquad R \to 0 \text{ as } \varepsilon \to 0 \text{ at fixed } (\sigma,\kappa).$$

The remainder R collects the smoothing-induced off-diagonal cross-terms ($p \neq q$) together with the higher prime-power contributions; it is not made uniform in κ in the present note. The constant term in the explicit formula vanishes by the antisymmetry of g^*.

The third is the renormalised analysis at the critical line (§§ 4–5). At $\sigma = \frac{1}{2}$ the auxiliary quantity $f_p = V_p(\frac{1}{2}) - 4(\log p)^2/p$ is shown to satisfy the closed form

$$f_p \;=\; \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \;>\; 0 \qquad (p \geq 2)$$

(Lemma 4.1), so that $c_p^{\mathrm{ren}} = f_p^{1/2}$ is real and strictly positive. The diagonal energy $D = \sum_p (c_p^{\mathrm{ren}})^2$ converges absolutely by comparison with $\sum_n (\log n)^2/n^2$ (Lemma 4.2) and evaluates to $D \approx 9.471$. The renormalised local curvature $H^{\mathrm{ren}}_{\mathrm{local}}(\frac{1}{2},\kappa) = H_{\mathrm{local}}(\frac{1}{2},\kappa) - 2(\log \kappa)^2$ is of lower order than $(\log \kappa)^2$ and exhibits a sawtooth structure: a positive jump f_p at each new prime $p \leq \kappa$, decreasing continuously between primes. The Mertens constant γ_M governs the average drift of this sawtooth via the classical estimate $\sum_{p \leq \kappa} (\log p)/p = \log \kappa + O(1)$.

The numerical comparison of § 6 establishes that $Z_{\mathrm{ren}} \geq H^{\mathrm{ren}}_{\mathrm{local}}$ holds at every point of the eight-point test grid except $\kappa = 20$, with the ratio growing from 1.5 at $\kappa = 29$ to over 10 at $\kappa = 1000$. Two open problems remain: the unconditional spectral inequality $Z(g^*_{1/2,\varepsilon}) \geq H_{\mathrm{local}}(\frac{1}{2},\kappa)$ (Problem 7.1) and the off-diagonal control of the cross-terms in the zero sum (Problem 7.2).

1.3 Reader Contract

This paper *proves*, by direct construction and classical identities, the admissibility of the test family (Lemma 2.1), the strict positivity of the renormalised weight $c_p^{\mathrm{ren}} = f_p^{1/2} > 0$

(Lemma 4.1), and the absolute convergence of the diagonal energy series $D = \sum_p (c_p^{\mathrm{ren}})^2$ (Lemma 4.2). These are unconditional mathematical statements.

This paper *establishes conditionally*, conditional on the standard remainder bookkeeping for the Weil explicit formula applied to antisymmetric Schwartz test functions, the prime-side identity for the Weil functional at fixed (σ, κ) as $\varepsilon \to 0$ (Proposition 3.1). The remainder is not made uniform in κ in the present note; the off-diagonal control needed for a uniform statement is the subject of Problem 7.2.

This paper *establishes numerically*, at reference parameters, the diagonal energy value $D \approx 9.471$ (Observation 4.3) and the finite-grid comparison $Z_{\mathrm{ren}} \geq H^{\mathrm{ren}}_{\mathrm{local}}$ on the eight-point grid $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$ at every tested point except $\kappa = 20$ (Observation 6.1). Both numerical statements are finite computations from the stated parameters and are labelled [numerical] in their theorem headings.

This paper *leaves open* two well-posed questions named in § 7: Problem 7.1 (Spectral inequality) and Problem 7.2 (Off-diagonal control). Both are formulated entirely within classical analytic number theory.

2 Construction of the admissible test family

Recall from [1, Eq. (3)] that

$$V_p(\sigma) \;=\; 4(\log p)^2 \, \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}.$$

Define the raw weight

$$c_{p,\sigma} \;:=\; V_p(\sigma)^{1/2} \cdot \frac{p^{1/4}}{(\log p)^{1/2}},$$

and let

$$\varphi_\varepsilon(x) \;=\; \frac{1}{\varepsilon\sqrt{2\pi}} \, e^{-x^2/(2\varepsilon^2)}$$

be the standard Gaussian bump. For fixed κ set

$$\widehat{f}_{\sigma,\varepsilon}(x) \;=\; \sum_{p \leq \kappa} c_{p,\sigma} \, \varphi_\varepsilon(x - \log p).$$

The admissible test function is obtained by antisymmetrisation:

$$\widehat{g}^*_{\sigma,\varepsilon}(x) \;:=\; \widehat{f}_{\sigma,\varepsilon}(x) \;-\; \widehat{f}_{\sigma,\varepsilon}(-x). \tag{1}$$

Lemma 2.1 (Admissibility; proved)**.** *For every $\sigma > 0$ and $\varepsilon > 0$, the function $g^*_{\sigma,\varepsilon}$ belongs to the admissible class S_{ad} used by Lagarias [3] in the formulation of Weil positivity.*

Proof. The Gaussian bumps φ_ε are Schwartz, and a finite linear combination of translated Schwartz functions is again Schwartz; antisymmetrisation preserves the Schwartz class. Hence $g^*_{\sigma,\varepsilon}$ and $\widehat{g}^*_{\sigma,\varepsilon}$ are rapidly decreasing. By antisymmetry, $\widehat{g}^*_{\sigma,\varepsilon}(0) = 0$ and $g^*_{\sigma,\varepsilon}(0) = 0$, providing the vanishing conditions at the origin required for admissibility. The Gaussian decay ensures absolute convergence of the prime-side and zero-side expressions appearing in the Weil functional for this family. □

3 The prime-side identity

Proposition 3.1 (Prime-side identity; conditional)**.** *For the family $g^*_{\sigma,\varepsilon}$ constructed in § 2,*

$$W(g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}) \;=\; Z(g^*_{\sigma,\varepsilon}) \;-\; H_{\mathrm{local}}(\sigma,\kappa) \;+\; R(\sigma,\varepsilon,\kappa), \tag{2}$$

where $Z(g^*_{\sigma,\varepsilon}) = \sum_\rho |\widehat{g}^*_{\sigma,\varepsilon}(\rho)|^2$ runs over the non-trivial zeros ρ of $\zeta(s)$, and the remainder $R(\sigma,\varepsilon,\kappa)$ tends to zero as $\varepsilon \to 0$ at fixed (σ,κ).

Sketch. The argument rests on the standard remainder bookkeeping for the Weil explicit formula applied to antisymmetric Schwartz test functions at fixed (σ,κ) as $\varepsilon \to 0$, which is the assumption under which the proposition is stated.

The Weil quadratic functional decomposes, in the Lagarias normalisation [3], into four pieces: a constant term, an archimedean integral, a prime sum, and a zero sum. The constant term, proportional to $\widehat{g}^*_{\sigma,\varepsilon}(0)^2$, vanishes by $\widehat{g}^*_{\sigma,\varepsilon}(0) = 0$ (Lemma 2.1). The archimedean integral, an integral against the Γ-factor of ξ, contributes $\psi_1(\sigma)$ in the limit $\varepsilon \to 0$ at fixed σ. The zero-side sum is $Z(g^*_{\sigma,\varepsilon})$ by definition. The prime-side quadratic form, expanded against the double sum in the convolution $g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}$, contributes

$$\sum_{p,q\le\kappa} c_{p,\sigma} c_{q,\sigma} \big[\varphi_\varepsilon * \check{\varphi}_\varepsilon\big](\log p - \log q) \cdot p^{-1/2} q^{-1/2}$$

together with prime-power terms of higher arithmetic order. By construction of the weights $c_{p,\sigma}$, the diagonal $p = q$ contribution converges to $\sum_{p\le\kappa} V_p(\sigma)$ as $\varepsilon \to 0$. The limit of the prime-side together with the archimedean integral therefore equals $H_{\mathrm{local}}(\sigma,\kappa)$. The remainder $R(\sigma,\varepsilon,\kappa)$ collects the smoothing-induced off-diagonal cross-terms ($p \ne q$) together with the higher prime-power contributions; both vanish as $\varepsilon \to 0$ at fixed (σ,κ) by Gaussian decay of $\varphi_\varepsilon * \check{\varphi}_\varepsilon$ at non-zero argument. The remainder is not made uniform in κ in the present note; the off-diagonal control needed for a uniform statement is the subject of Problem 7.2 of § 7. □

Remark. *Equation* (2) *is the bridge between the curvature field of [1] and the Weil-positivity framework of [3]. The vanishing of the constant term, by condition (S2) on the admissible class, is the structural reason that the right-hand side of* (2) *carries no free additive term: the archimedean integral does not vanish but contributes $\psi_1(\sigma)$, which is then absorbed into $H_{\mathrm{local}}(\sigma,\kappa)$.*

4 Renormalised weights and convergence

At $\sigma = \frac{1}{2}$ define

$$f_p \;:=\; V_p(\tfrac{1}{2}) \;-\; \frac{4(\log p)^2}{p}.$$

A direct calculation gives

$$f_p \;=\; \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \;>\; 0 \qquad (p \geq 2). \tag{3}$$

Lemma 4.1 (Positivity of the renormalised weight; proved)**.** *For every prime $p \geq 2$, the quantity f_p in (3) is strictly positive.*

Proof. The numerator $2p-1 > 0$ and the denominator $p(p-1)^2 > 0$ for every prime $p \geq 2$. □

The renormalised weight is then simply

$$c_p^{\mathrm{ren}} \;:=\; f_p^{1/2} \;>\; 0, \tag{4}$$

so that $(c_p^{\mathrm{ren}})^2 = f_p$ exactly. Numerical values for small primes:

p	$V_p(\frac{1}{2})$	$4(\log p)^2/p$	$f_p = (c_p^{\mathrm{ren}})^2$
2	3.844	0.961	2.883
3	3.621	1.609	2.012
5	3.238	2.072	1.166
7	2.945	2.164	0.781
11	2.530	2.091	0.439

Lemma 4.2 (Convergence of the diagonal energy; proved)**.** *The series*

$$D \;:=\; \sum_p (c_p^{\mathrm{ren}})^2 \;=\; \sum_p \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \tag{5}$$

converges absolutely.

Proof. For large p, $(c_p^{\mathrm{ren}})^2 \;\sim\; 8(\log p)^2/p^2$. Since $\sum_n (\log n)^2/n^2 \;<\; \infty$, absolute convergence follows by comparison with this dominating series. □

Observation 4.3 (Diagonal energy value; numerical)**.** *Numerically,* $D \approx 9.471$ *when summed over primes below* 10^4*, with tail bound below* 0.009*.*

5 The renormalised scale and the Mertens constant

By the prime number theorem,

$$H_{\mathrm{local}}(\tfrac{1}{2}, \kappa) \;\sim\; 2(\log \kappa)^2 \qquad (\kappa \to \infty),$$

so the renormalised remainder

$$H_{\mathrm{local}}^{\mathrm{ren}}(\tfrac{1}{2},\kappa) \;:=\; H_{\mathrm{local}}(\tfrac{1}{2},\kappa) \;-\; 2(\log\kappa)^2$$

is of lower order than $(\log\kappa)^2$. Numerically, $H_{\mathrm{local}}^{\mathrm{ren}}$ exhibits a sawtooth pattern: at each new prime $p \le \kappa$ it jumps upward by $f_p > 0$, and decreases continuously between primes as $2(\log\kappa)^2$ grows. This is the direct analogue of the prime-counting function $\pi(x)$, which increases by 1 at each prime and is constant between consecutive primes. The Mertens constant γ_M governs the average drift of this sawtooth via the classical estimate

$$\sum_{p\le\kappa} \frac{\log p}{p} \;=\; \log\kappa + O(1) \qquad (\kappa\to\infty),$$

connecting the renormalised scale to classical prime-sum estimates [6].

Remark. *The Mertens constant*

$$\gamma_M \;=\; \gamma + \sum_p \Big[\log\big(1-\tfrac{1}{p}\big) + \tfrac{1}{p}\Big] \;\approx\; 0.2615,$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, appears in prime-sum estimates that control the average behaviour of $H_{\mathrm{local}}^{\mathrm{ren}}$. The same constant γ enters the first Li coefficient [5]

$$\lambda_1 \;=\; 1 + \tfrac{\gamma}{2} - \log 2 - \tfrac{\log\pi}{2} \;\approx\; 0.0231,$$

reflecting the universal role of γ in the transition from discrete prime sums to continuous logarithmic integrals.

Remark. *The renormalised construction clarifies the scale of the prime-side contribution but does not by itself resolve the spectral inequality of Problem 7.1.*

6 Numerical comparison of zero and prime sides

We compare the renormalised zero sum Z_{ren} with the renormalised local curvature $H_{\mathrm{local}}^{\mathrm{ren}}$ on a finite grid in κ. The smoothing parameter $\varepsilon = 0.05$ is required for meaningful numerical experiments with the zero sum: for $\varepsilon = 0.5$ one has $e^{-\varepsilon^2\gamma_1^2} \approx 10^{-22}$, effectively suppressing all zero contributions, so that no observable signal remains. With $c_p^{\mathrm{ren}} = f_p^{1/2}$, $\varepsilon = 0.05$, and $N = 100$ zeros:

κ	Z_{ren}	$H^{\text{ren}}_{\text{local}}$	$Z_{\text{ren}} - H^{\text{ren}}_{\text{local}}$	$Z_{\text{ren}} \geq H^{\text{ren}}_{\text{local}}$
10	4.888	3.044	+1.844	✓
20	3.782	4.770	−0.988	×
29	5.520	3.588	+1.931	✓
50	5.693	2.881	+2.812	✓
100	5.311	1.562	+3.748	✓
200	5.102	2.355	+2.747	✓
500	5.156	1.101	+4.056	✓
1000	4.289	0.426	+3.863	✓

Observation 6.1 (Finite-grid stability of the comparison; numerical)**.** *On the eight-point grid $\kappa \in \{10, 20, 29, 50, 100, 200, 500, 1000\}$, the inequality $Z_{\text{ren}} \geq H^{\text{ren}}_{\text{local}}$ holds at every tested point except $\kappa = 20$, where $Z_{\text{ren}} - H^{\text{ren}}_{\text{local}} = -0.988$. The ratio $Z_{\text{ren}}/H^{\text{ren}}_{\text{local}}$ grows from 1.5 at $\kappa = 29$ to over 10 at $\kappa = 1000$. No claim is made about κ outside the eight tested points.*

The single grid violation $Z_{\text{ren}} < H^{\text{ren}}_{\text{local}}$ at $\kappa = 20$ illustrates the sawtooth behaviour of $H^{\text{ren}}_{\text{local}}$: directly after a prime jump $H^{\text{ren}}_{\text{local}}$ rises by f_p, then decreases continuously until the next prime, exactly as $\pi(x) - \text{Li}(x)$ fluctuates near small primes before stabilising asymptotically. The pattern observed on the grid for $\kappa \geq 29$ is consistent with the spectral inequality of Problem 7.1, but does not prove it: the underlying object is the unconditional positivity of the zero-side quadratic form for an admissible test function in S_{ad}, which is not implied by finite-grid evaluations.

Remark (The constant $D/2\pi$)**.** *The formal asymptotic $Z(g^*) \sim D \log\kappa \log\log\kappa$, derivable under unconditional control of the off-diagonal cross-terms (Problem 7.2), yields a natural bridge constant*

$$\frac{D}{2\pi} \approx \frac{9.471}{6.283} \approx 1.507,$$

measuring the ratio of the prime-side diagonal energy D to the archimedean normalisation 2π of the zero-counting density $N(T) \sim T \log T/(2\pi)$. This constant is of a different nature from the pair-correlation statistics studied in the GUE conjecture [7]: it depends only on the total density of zeros via the classical formula for $N(T)$, which is unconditional and requires neither GUE nor RH as input. Whether $D/2\pi$ admits a representation in terms of classical constants is left open.

All numerical claims of this section are reproducible from the verification scripts in the GitHub repository [2].

7 Open Problems

The present work reduces the question of prime-side derivation of Weil positivity for the family $g^*_{\sigma,\varepsilon}$ to two well-posed subproblems. We name each and indicate the tools we expect to be relevant.

Open Problem 7.1 (Spectral inequality on the critical line)**.** *Prove that for every cutoff κ and every sufficiently small $\varepsilon > 0$,*

$$Z(g^*_{1/2,\varepsilon}) \;\geq\; H_{\mathrm{local}}(\tfrac{1}{2}, \kappa). \tag{6}$$

This inequality is of RH strength. By Lagarias [3], the equivalence RH $\iff W(f * \check{f}) \geq 0$ *holds for all $f \in S_{\mathrm{ad}}$. A positive answer to* (6) *establishes $W(g^*_{1/2,\varepsilon} * \check{g}^*_{1/2,\varepsilon}) \geq 0$ for the specific family constructed in § 2, which is necessary, but not sufficient, for RH in the absence of a density argument for this family in S_{ad}. The expected tools are: explicit lower bounds on the zero-side quadratic form by the unconditional zero-counting density; control of the cross-terms in the zero sum (Problem 7.2). On the tested grid the inequality $Z_{\mathrm{ren}} \geq H^{\mathrm{ren}}_{\mathrm{local}}$ holds at every tested point except $\kappa = 20$ (Observation 6.1), consistent with* (6) *but not establishing it analytically.*

Open Problem 7.2 (Off-diagonal control of the zero sum)**.** *Prove unconditionally that the off-diagonal contribution*

$$\sum_{\substack{p,q\leq\kappa \\ p\neq q}} c_p^{\mathrm{ren}}\, c_q^{\mathrm{ren}} \sum_{\rho} \widehat{g}^*_{1/2,\varepsilon}(\rho)\, \overline{\widehat{g}^*_{1/2,\varepsilon}(\rho)}\,\Big|_{p\neq q \text{ part}}$$

is asymptotically negligible compared with the diagonal contribution

$$\sum_p (c_p^{\mathrm{ren}})^2 \cdot I(\varepsilon) \;\sim\; D\,\log\kappa,$$

so that, combined with the unconditional density $N(T) \sim T\log T/(2\pi)$, the formal asymptotic $Z(g^*) \sim D\log\kappa\log\log\kappa$, expected on the Selberg-orthogonality heuristic for independent zero contributions [8], becomes a proved statement. The diagonal part is unconditional: it follows from the prime number theorem and the absolute convergence of (5)*. The expected tools are: analytic continuation of $\sum_\gamma \cos(\gamma\log(p/q))$ for $p \neq q$ via classical explicit-formula techniques; cancellation by oscillation, with the $(\log\kappa)^{-1}$-decay of the diagonal entering as a size constraint. This problem is formulated entirely within classical analytic number theory.*

8 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The non-trivial zeros ρ enter the construction only through the classical zero-counting density $N(T) \sim T\log T/(2\pi)$, which is unconditional, and through finite Gaussian-smoothed sums for which the symmetric pairing $\rho \leftrightarrow \bar{\rho}$ is used without further hypothesis. The reference point $\sigma = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The construction of $g^*_{\sigma,\varepsilon}$ uses only Schwartz-class smoothing and antisymmetrisation; it makes no reference to random-matrix statistics, pair-correlation conjectures, or any postulated spectral realisation of the zeros. The bridge constant $D/2\pi$ identified in § 6 depends only on the total density of zeros via the classical formula for $N(T)$ and is structurally independent of GUE.

The Weil explicit formula is classical. Proposition 3.1 applies the Weil explicit formula [4] in the Lagarias admissible formulation [3]; both are valid for every sufficiently regular test function and require no positivity hypothesis beyond the classical ones used in the derivation of the explicit formula itself.

Numerical results are clearly marked. The diagonal energy value $D \approx 9.471$ and the comparison-grid statements of § 6 rely on direct computation; each statement is labelled [numerical] in its heading.

Open problems are named, not used as hypotheses. Problems 7.1 and 7.2 are stated openly in § 7. No claim of this paper is conditional on a positive answer to either problem.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

The author welcomes critical feedback, corrections, and collaboration: Zenodo (`https://zenodo.org/search?q=Ulrich+Tehrani`) and GitHub (`https://github.com/utehrani/analysislab-nt`).

Topics of particular interest include: analytical approaches to the spectral inequality of Problem 7.1; unconditional control of the off-diagonal cross-terms of Problem 7.2; and the relation between the bridge constant $D/2\pi$ identified in § 6 and classical analytic constants attached to the zero-counting density.

We particularly welcome expert comment on the following question: for the Gaussian-smoothed admissible family $g^*_{\sigma,\varepsilon}$ constructed in § 2, is the comparison $Z(g^*) \sim D \log \kappa \log \log \kappa$, with D as in (5), accessible by classical sieve or explicit-formula techniques without recourse to random-matrix statistics?

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